



Chapter 5 Discrete Fourier Transform

—— DFT



Outline



- **Orthogonal Transforms**
 - ✓ **The Definition of DFT**
 - ✓ **The Relationship Between DFT and DTFT**
 - ✓ **The Computation of Finite-Length Sequences**
- **Classifications of Finite-Length Sequences**
 - ✓ **DFT Properties and Theorems**
- **Fourier-Domain Filtering**
- **Computation of the DFT of Real sequences**
 - ✓ **Linear Convolution Using DFT**

5.1 Orthogonal Transforms



- General form of the orthogonal transform pair:

$$X[k] = \sum_{n=0}^{N-1} x[n] \psi^*[k, n] \quad 0 \leq k \leq N-1$$

analysis equation

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \psi[k, n] \quad 0 \leq n \leq N-1$$

synthesis equation

- $\psi[k, n]$ --- basis sequences, are also length-N sequences in both domains, which satisfy:

$$\frac{1}{N} \sum_{n=0}^{N-1} \psi[k, n] \psi^*[l, n] = \begin{cases} 1, l = k \\ 0, l \neq k \end{cases}$$

A periodic sequence $\tilde{x}[n] = \tilde{x}[n + rN]$ (r is integer, N is period) can be represented as a **Fourier series**.

The orthogonal functions set is: $\{\psi[k, n] = e^{j\left(\frac{2\pi}{N}\right)kn}\} \quad (k = 0, 1, 2, \dots, N - 1)$

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi}{N}kn} = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn} \quad \dots(\text{A})$$

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn} \quad \dots(\text{B})$$

$$W_N = e^{-j\frac{2\pi}{N}} \quad (\text{kernel})$$



Note:

① $\psi[k, n]$ has period N , $\psi[0, n] = 1$ is DC component, $\psi[k, n]$ is the k th-harmonic.

$$\textcircled{2} \quad \sum_{n=0}^{N-1} \psi[k, n] \psi^*[\ell, n] = \sum_{n=0}^{N-1} e^{j \frac{2\pi}{N} (k-\ell)n} = \frac{1 - e^{j \frac{2\pi}{N} (k-\ell)N}}{1 - e^{j \frac{2\pi}{N} (k-\ell)}} = \begin{cases} 0, & \ell \neq k \\ N, & \ell = k + rN \end{cases}$$

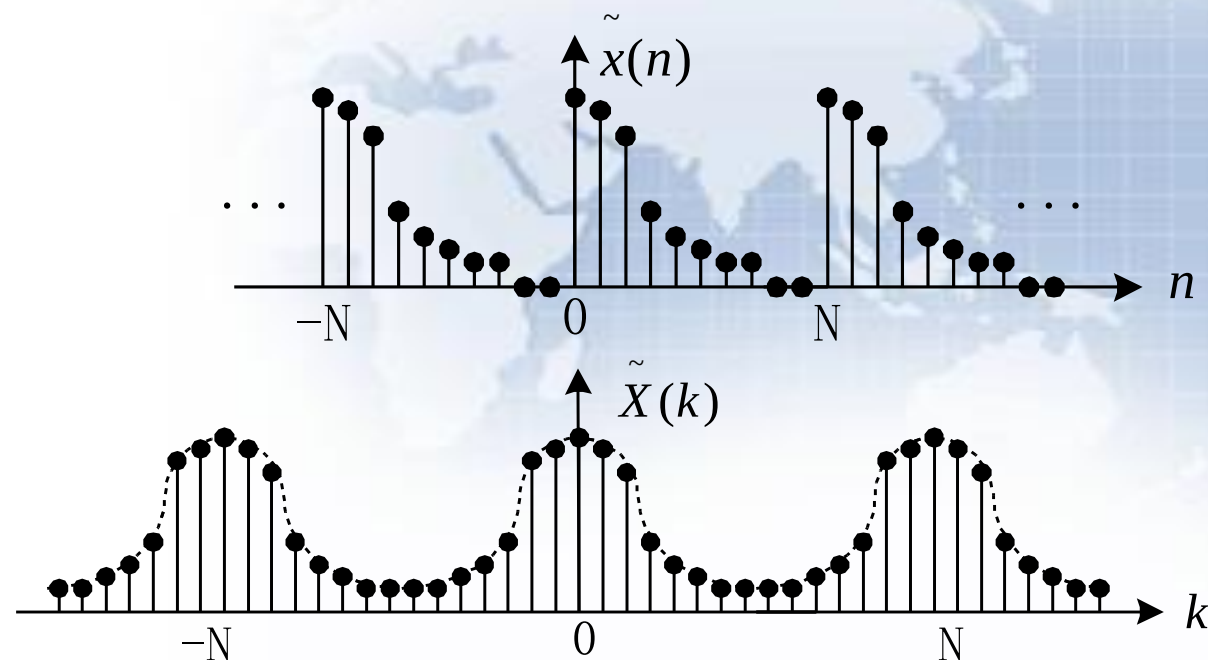
$$\textcircled{3} \quad \psi[k + N, n] = \psi[k, n] \quad \left\{ \psi[k, n] = e^{j \left(\frac{2\pi}{N} \right) kn} \right\}$$

④ The equation (A) and (B) are expressed as



$$\tilde{x}[n] \Leftrightarrow \tilde{X}[k]$$

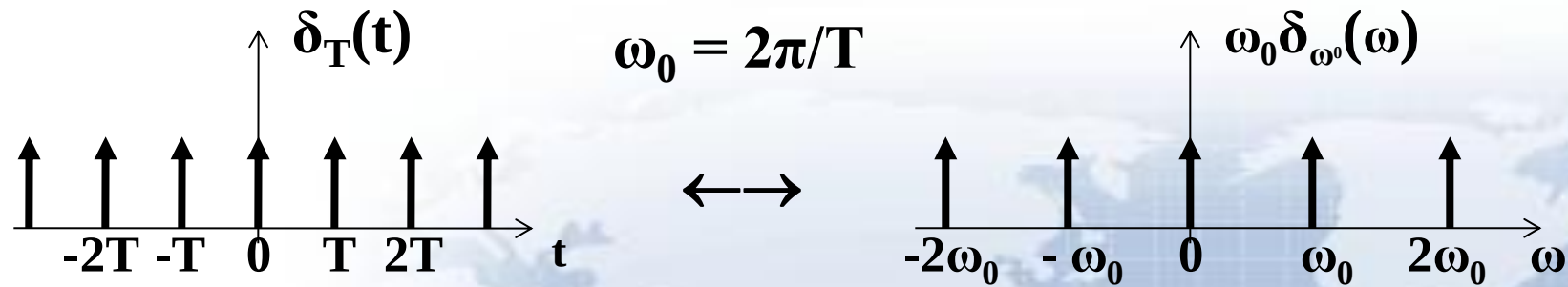
or $\tilde{X}[k] = DFS\{\tilde{x}[n]\} \quad \tilde{x}[n] = IDFS\{\tilde{X}[k]\}$



Typical DFS Pair



- $\delta_T(t) \longleftrightarrow \omega_0 \delta_{\omega_0}(\omega)$



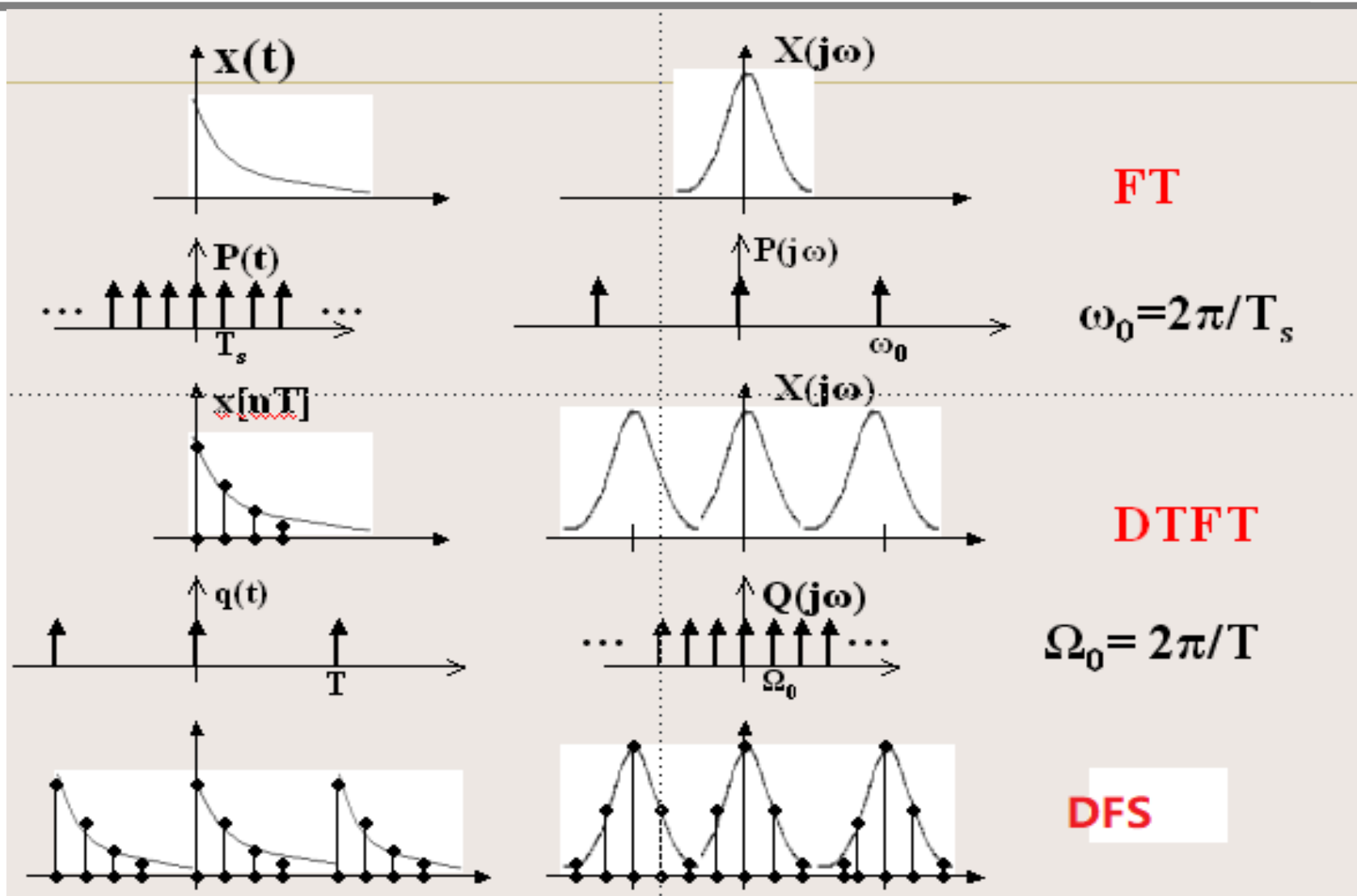
- The signals in both sides are **periodical**, so the processing could be in **one period**, which is important because:
 - (1) the number of calculation is limited, which is necessary for computer;
 - (2) all of the signal information could be kept in one period, which is necessary for accurate processing.

Make a Signal Discrete and Periodical



- The engineering signals are often continuous and aperiodical. If we want to process the signals with DFS, we have to make the signals discrete and periodical.
- Sampling to make the signal be discrete.
- Make the signal periodical by periodical expanding.
 - If $x[n]$ is a **limited length** N-point sequence, see it as one period of a periodical signal;
 - If $x[n]$ is an infinite length sequence, **cut-off** its tail to make a N-point sequence, then do the periodic extending. The **truncation** will introduce distortion. ---- windowing

Make a signal discrete and periodical



5.2 The Definition of DFT



$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, 0 \leq k \leq N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, 0 \leq n \leq N-1$$

Where: $W_N = e^{-j2\pi/N}$

$$x[n] \overset{DFT}{\underset{N}{\longleftrightarrow}} X[k]$$

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Verify:



$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn} = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{l=0}^{N-1} X[l] W_N^{-ln} \right) W_N^{kn} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} X[l] W_N^{(k-l)n} = \sum_{l=0}^{N-1} X[l] \left\{ \frac{1}{N} \sum_{n=0}^{N-1} W_N^{(k-l)n} \right\} \end{aligned}$$

$$\because \sum_{n=0}^{N-1} W_N^{(k-l)n} = \begin{cases} N, & \text{for } k-l = mN, \text{ } m \text{ an integer} \\ 0, & \text{otherwise} \end{cases}$$

Hence,
$$X[k] = \sum_{l=0}^{N-1} X[l] \delta[k - \ell]$$

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Example: Consider the length- N sequence



$$x[n] = \begin{cases} 1, & n = 0 \\ 0, & 1 \leq n \leq N-1 \end{cases}$$

Its N -point DFT is given by:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} = x[0] W_N^0 = 1$$
$$0 \leq k \leq N-1$$

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Example: Consider the length- N sequence



$$g[n] = \cos(2\pi rn / N), \quad 0 \leq r \leq N - 1$$

• Using a trigonometric identity we can write:

$$g[n] = \frac{1}{2} \left(e^{j2\pi rn / N} + e^{-j2\pi rn / N} \right) = \frac{1}{2} \left(W_N^{-rn} + W_N^{rn} \right)$$

$$G[k] = \sum_{n=0}^{N-1} g[n] W_N^{kn} = \frac{1}{2} \left(\sum_{n=0}^{N-1} W_N^{-(r-k)n} + \sum_{n=0}^{N-1} W_N^{(r+k)n} \right),$$

$$0 \leq k \leq N - 1$$

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Making use of the identity:

$$\sum_{n=0}^{N-1} W_N^{-(k-\ell)n} = \begin{cases} N, & \text{for } k - \ell = mN, \text{ } m \text{ an integer} \\ 0, & \text{otherwise} \end{cases}$$

we get:

$$G[k] = \begin{cases} N / 2, & \text{for } k = r \\ N / 2, & \text{for } k = N - r \\ 0, & \text{otherwise} \end{cases} \quad 0 \leq k \leq N - 1$$

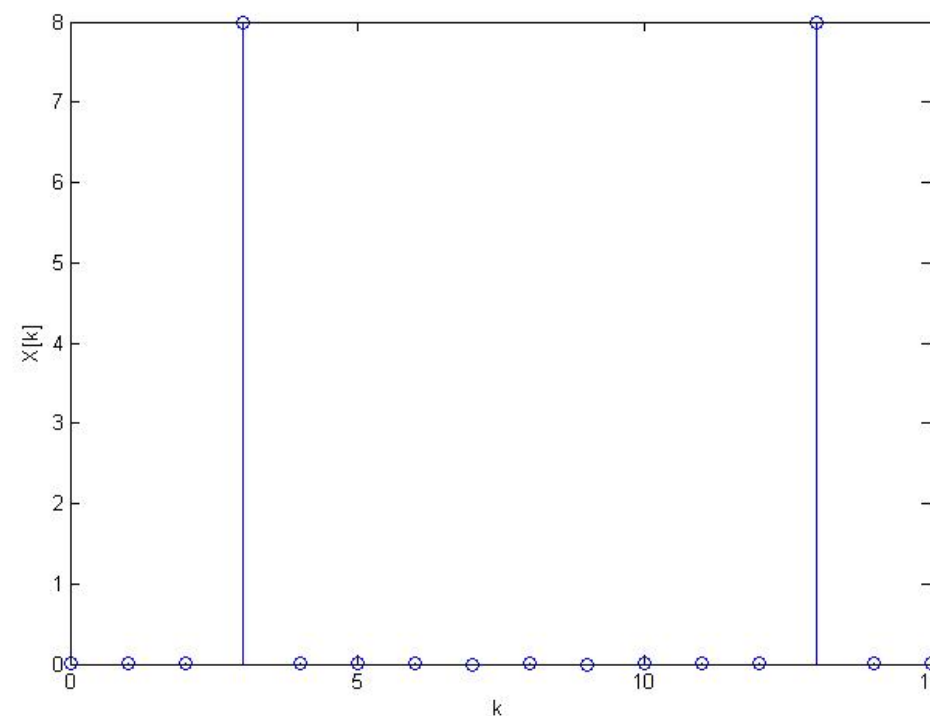
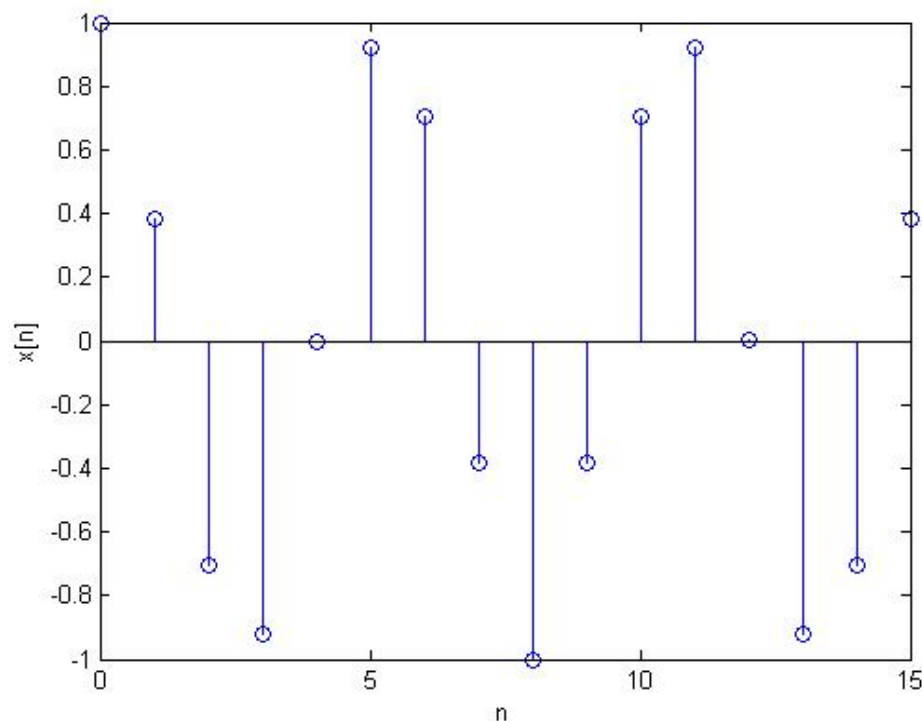
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- When $N=16$, $r=3$



$$g[n] = \cos(6\pi n/16), \quad 0 \leq n \leq 15$$

$$G[k], 0 \leq k \leq 15$$



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Matrix Relations



- The DFT samples defined by:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1$$

Can be expressed in matrix form as $\mathbf{X} = \mathbf{D}_N \mathbf{x}$

Where,

$$\mathbf{X} = [x[0] \quad x[1] \quad \dots \quad x[N-1]]^T$$

$$\mathbf{x} = [x[0] \quad x[1] \quad \dots \quad x[N-1]]^T$$

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Matrix Relations



- And D_N is the $N \times N$ DFT matrix given by:

$$D_N = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N^1 & W_N^2 & \cdots & W_N^{(N-1)} \\ 1 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & W_N^{(N-1)} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)^2} \end{bmatrix}$$

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Matrix Relations



- Likewise, the IDFT relation is given by:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq k \leq N-1$$

can be expressed in matrix form as $\mathbf{x} = \mathbf{D}_N^{-1} \mathbf{X}$,
where $\mathbf{D}_N^{-1}$ is the $N \times N$ IDFT matrix.

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Matrix Relations



$$\mathbf{D}_N^{-1} = \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N^{-1} & W_N^{-2} & \dots & W_N^{-(N-1)} \\ 1 & W_N^{-2} & W_N^{-4} & \dots & W_N^{-2(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & W_N^{-(N-1)} & W_N^{-2(N-1)} & \dots & W_N^{-(N-1)^2} \end{bmatrix}$$

Note: $\mathbf{D}_N^{-1} = (\mathbf{D}_N^*) / N$

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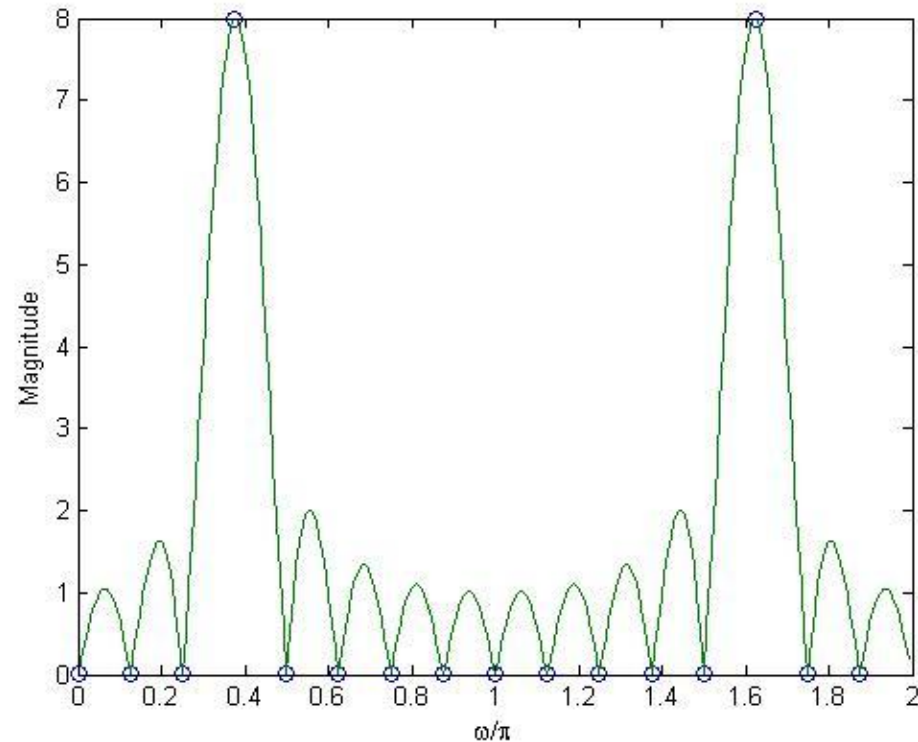
DFT Computation Using MATLAB



- The functions to compute the DFT and the IDFT are **FFT** and **IFFT**.
- These functions make use of FFT algorithms which are computationally highly efficient compared to the direct computation.
 - From $O(n^2)$ to $O(n\log_2 n)$
- **Programs 5_1.m** and **5_2.m** illustrate the use of these functions.

- Example - **Program 5_3.m** can be used to compute the DFT and the DTFT of the sequence:

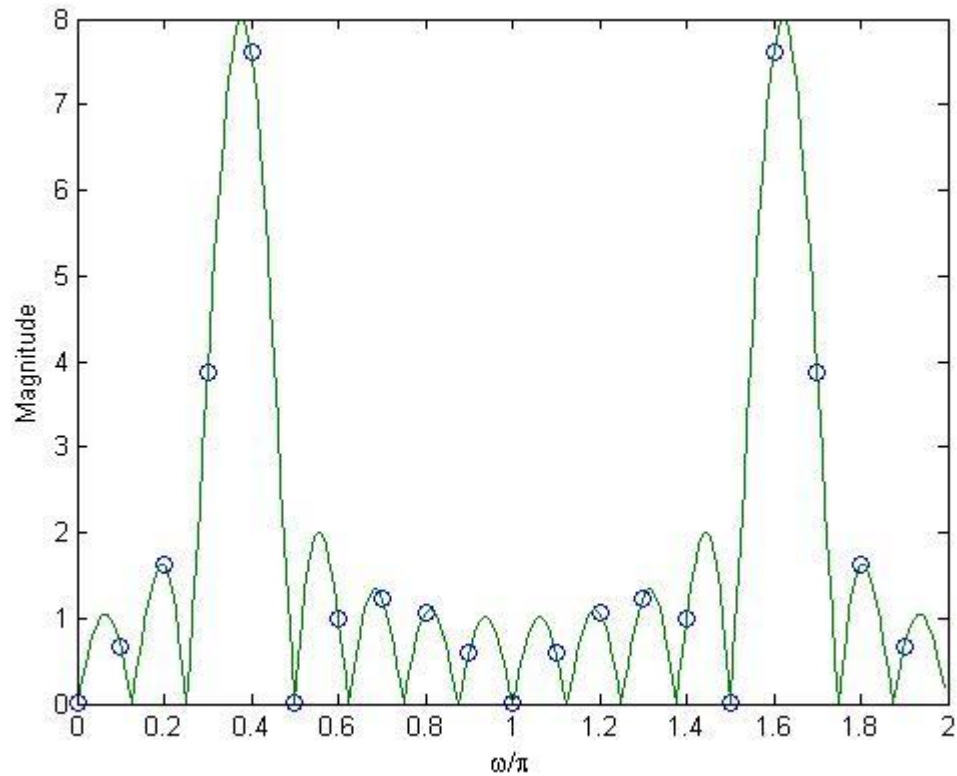
$$x[n] = \cos(6\pi n/16), \quad 0 \leq n \leq 15$$



•fft(x, 16)

○ indicates DFT samples

- $\text{fft}(x, 20)$



Why?



5.3 The Relationship Between DFT and DTFT

- *From DTFT to DFT*
- *DTFT from DFT by Interpolation*
- *Sampling the DTFT*



From DTFT to DFT

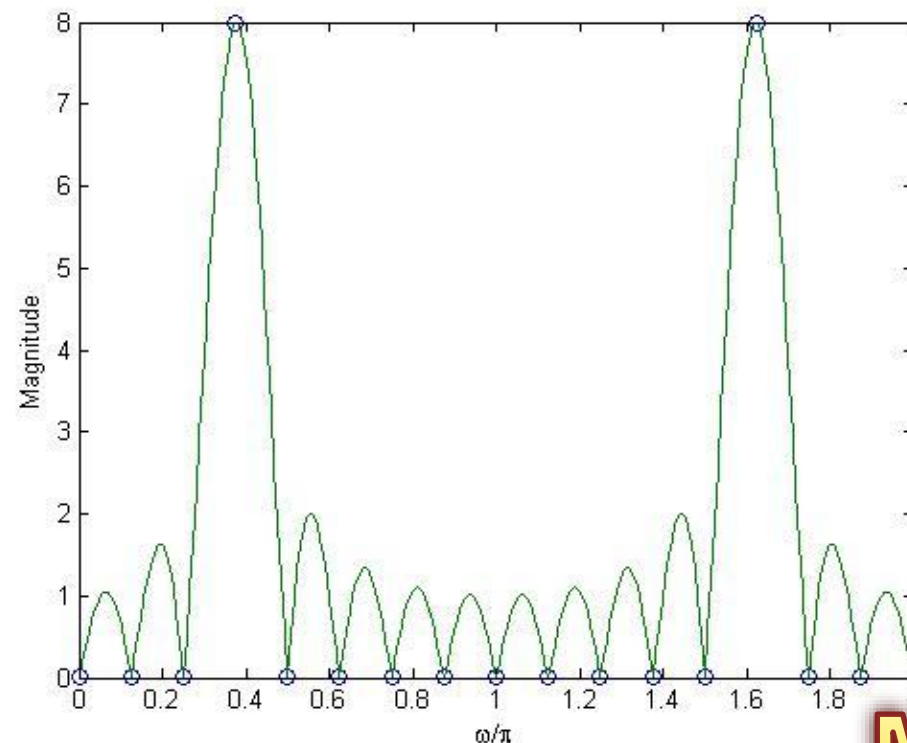


- Sampling the DTFT of a N points sequence with the space is $2\pi/N$ to get N frequency points, that is DFT.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}$$

$$X[k] = X(e^{j\omega})|_{\omega=2\pi k/N} = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$$

$$0 \leq \omega < 2\pi, 0 \leq k \leq N-1$$

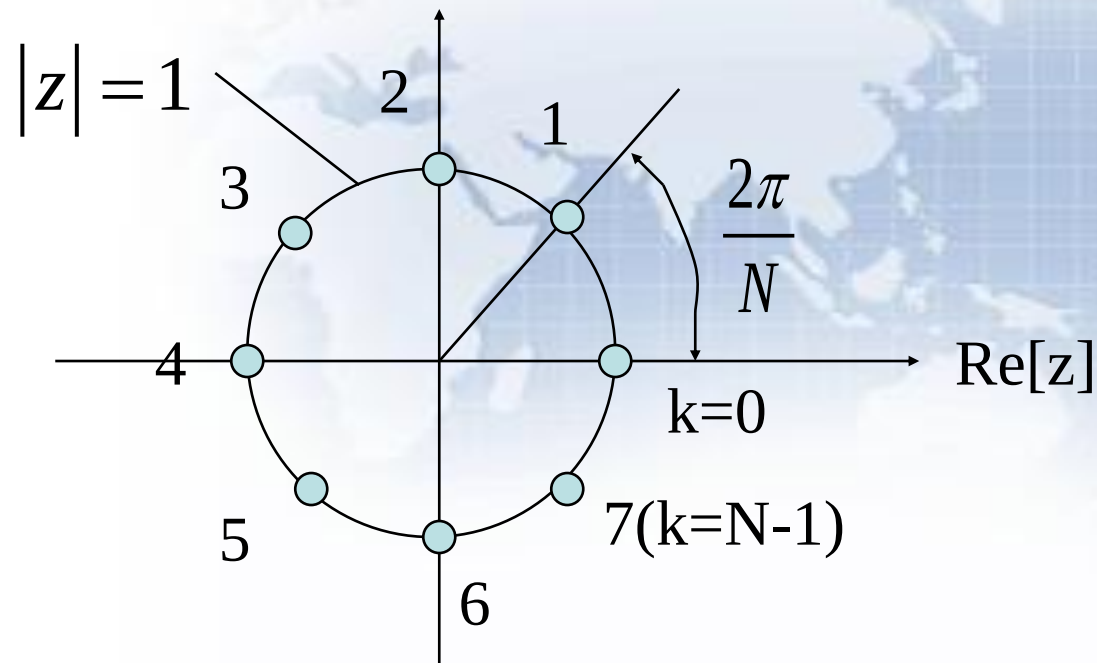


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From DTFT to DFT



- **DTFT: discrete in time domain, continuous in frequency domain.**
- **Sampling the DTFT of sequences to get N frequency points to research, that is DFT.**



Numerical Computation of the DTFT Using the DFT



- How to compute the DTFT of a length- N sequence $x[n]$ by using computer?
 1. Sampling the $X(e^{j\omega})$ to get a length- M sequence, $M \gg N$. Each frequency component is $X(e^{j2\pi/M})$.
 2. If the DFT of a sequence is length- M , the sequence must be a length- M sequence. So we build a length- M sequence use $x[n]$.
- The function **freqz** employs this approach to evaluate the frequency response at a prescribed set of frequencies of a DTFT expressed as a rational function in $e^{-j\omega}$.

$$x_e[n] = x[n], 0 \leq n \leq N-1$$

$$x_e[n] = 0, N \leq n \leq M-1$$

$$X(e^{j\omega_k}) = \sum_{n=0}^{M-1} x_e[n] e^{-j2\pi kn/M}$$

- We can see above equation is the DFT of a length-M sequence. Because $M \gg N$, it can be seen as an approach of the DTFT of a length-N sequence.

Sampling the DTFT



- Consider a sequence $x[n]$ with DTFT $X(e^{j\omega})$
- We sample $X(e^{j\omega})$ at N equally spaced points $\omega_k = 2\pi k/N$, $0 \leq k \leq N-1$ developing the N frequency samples $\{X(e^{j\omega_k})\}$
- These N frequency samples can be considered as an N -point DFT $Y[k]$ whose N -point IDFT is a length- N sequence $y[n]$.

What's relationship between $y[n]$ and $x[n]$?

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Sampling the DTFT



$$X(e^{j\omega}) = \sum_{\ell=-\infty}^{\infty} x[\ell]e^{-j\omega\ell}$$

Thus: $Y[k] = X(e^{j\omega_k}) = X(e^{j2\pi k/N})$

$$= \sum_{l=-\infty}^{\infty} x[l]e^{-j2\pi kl/N} = \sum_{l=-\infty}^{\infty} x[l]W_N^{kl}$$

IDFT of $Y[k]$ yields:

$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} Y[k]W_N^{-kn}, 0 \leq n \leq N-1$$

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Sampling the DTFT



$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{l=-\infty}^{\infty} x[l] W_N^{kl} W_N^{-kn} = \sum_{l=-\infty}^{\infty} x[l] \left\{ \frac{1}{N} \sum_{k=0}^{N-1} W_N^{-k(n-l)} \right\}$$

Making use of the identity:

$$\frac{1}{N} \sum_{n=0}^{N-1} W_N^{-k(n-l)} = \begin{cases} 1, & \text{for } l - n = mN \\ 0, & \text{otherwise} \end{cases} = \sum_{m=-\infty}^{\infty} \delta[n - l + mN]$$

We have: $y[n] = \sum_{m=-\infty}^{\infty} x[n + mN], \quad 0 \leq n \leq N-1$

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Sampling the DTFT



- **To apply:**
$$y[n] = \sum_{m=-\infty}^{\infty} x[n + mN], \quad 0 \leq n \leq N - 1$$

to finite-length sequences.

- If $x[n]$ is a length- M sequence with $M \leq N$, then $y[n] = x[n]$ for $0 \leq n \leq N-1$.
- If $M > N$, there is a **time-domain aliasing** of samples of $x[n]$ in generating $y[n]$, and $x[n]$ cannot be recovered from $y[n]$.

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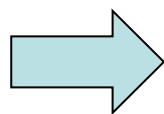
- Example: Let $\{x[n]\}=\{0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5\}$



By sampling its DTFT $X(e^{j\omega})$ at $\omega_k=2\pi k/4$, $0 \leq k \leq 3$ and then applying a 4-point IDFT to these samples, we arrive at the sequence $y[n]$ given by:

$$y[n] = \sum_{m=-\infty}^{\infty} x[n+4m], \quad 0 \leq n \leq N-1$$

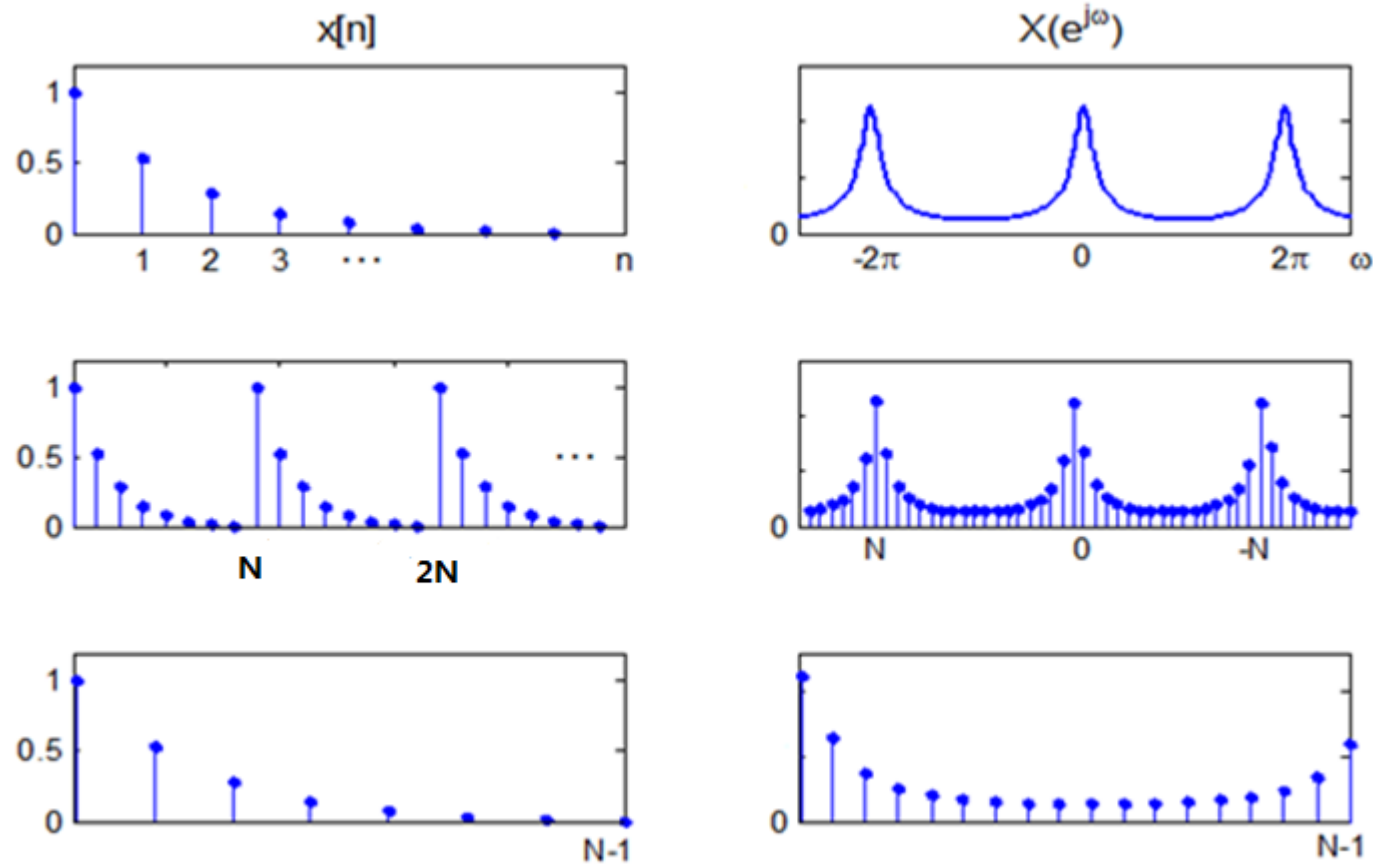
- i.e. $\{y[n]\}=\{4 \quad 6 \quad 2 \quad 3\}$



$\{x[n]\}$ cannot be recovered from $\{y[n]\}$

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Processing a signal in discrete frequency



DTFT

DFS

DFT

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Make a signal discrete and periodical



Time domain

Frequency domain

Continuous Aperiodical $\leftarrow$ **CTFT** $\rightarrow$ Continuous Aperiodical

Continuous Periodical $\leftarrow$ **FS** $\rightarrow$ Aperiodical Discrete

Discrete Aperiodical $\leftarrow$ **DTFT** $\rightarrow$ Periodical Continuous

Discrete & Periodical $\leftarrow$ **DFS** $\rightarrow$ Periodical & Discrete

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DTFT from DFT by Interpolation



- Given the N -point DFT $X[k]$ of a length- N sequence $x[n]$, its DTFT $X(e^{j\omega})$ can be uniquely determined from $X[k]$.

How?

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DTFT from DFT by Interpolation



$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{N-1} x[n] e^{-j\omega n} \\ &= \sum_{n=0}^{N-1} \left\{ \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \right\} e^{-j\omega n} \\ &= \sum_{k=0}^{N-1} X[k] \underbrace{\left\{ \frac{1}{N} \sum_{n=0}^{N-1} e^{-j(\omega - \frac{2\pi k}{N})n} \right\}}_{\Phi_k(\omega)} \end{aligned}$$

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DTFT from DFT by Interpolation



$$\Phi_k(\omega) = \frac{1}{N} \frac{1 - e^{-j(\omega - \frac{2\pi k}{N})N}}{1 - e^{-j(\omega - \frac{2\pi k}{N})}} = \frac{1}{N} \frac{\sin(\frac{\omega N - 2\pi k}{2})}{\sin(\frac{\omega N - 2\pi k}{2N})} e^{-j[(\omega - 2\pi k/N)][(N-1)/2]}$$

- Therefore:

$$X(e^{j\omega}) = \sum_{k=0}^{N-1} X[k] \cdot \Phi_k(\omega)$$

- It is **interpolated** in frequency domain.

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5.4 The Computation of Finite-Length Sequences

- *Circular Shift*
- *Circular reversal*
- *Circular convolution*



Circular Shift



- **Circular Shift** of length-N sequence $x[n]$:

$$x[\langle n - n_0 \rangle_N] = \begin{cases} x[n - n_0], & \text{for } n_0 \leq n \leq N - 1 \\ x[N - n_0 + n], & \text{for } 0 \leq n \leq n_0 \end{cases}$$

$\langle \bullet \rangle_N$ — modulo operation

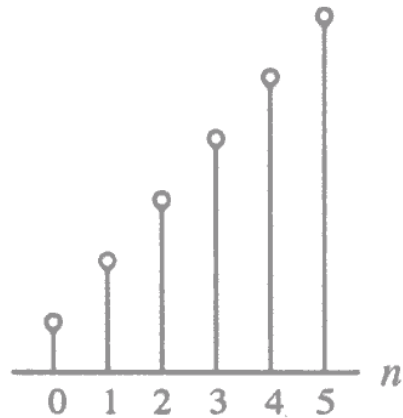
$$\langle -1 \rangle_6 = \langle 6 - 1 \rangle_6 = \langle 5 \rangle_6$$

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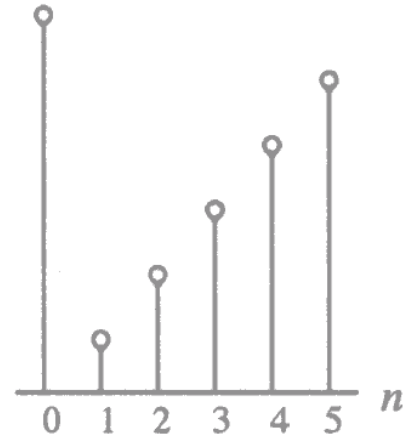
Circular Shift



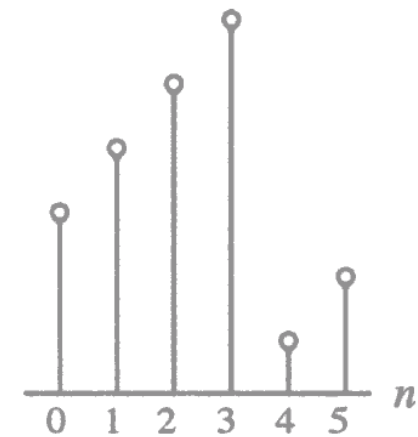
Illustration of the concept of a circular shift:



$$x[n]$$



$$\begin{aligned} & x[\langle n - 1 \rangle_6] \\ &= x[\langle n + 5 \rangle_6] \end{aligned}$$



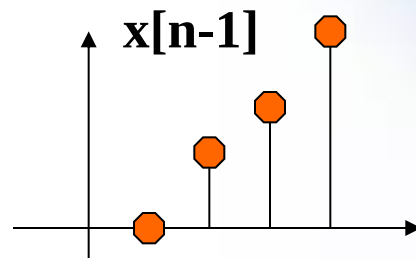
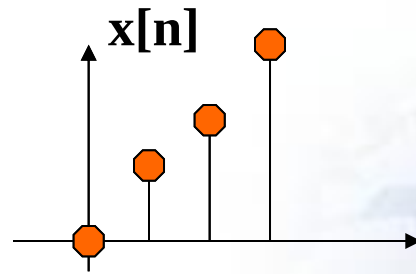
$$\begin{aligned} & x[\langle n - 4 \rangle_6] \\ &= x[\langle n + 2 \rangle_6] \end{aligned}$$

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Circular Shift of a Sequence



Non-circular shift



circular shift



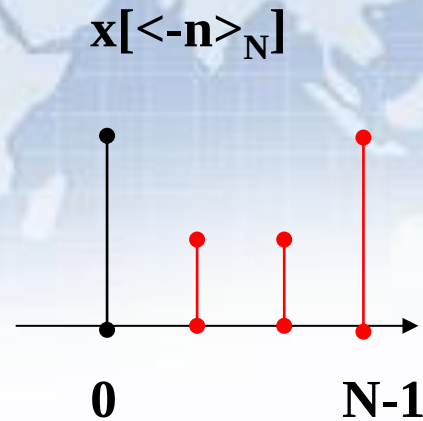
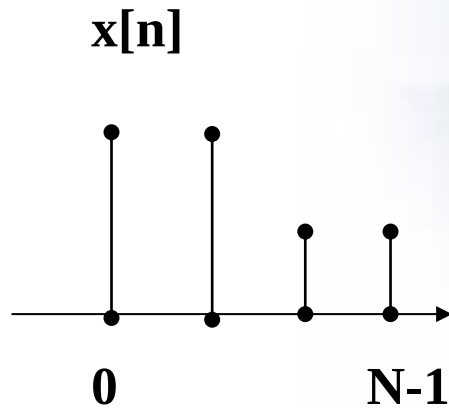
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Circular reversal



- **Circular reversal** of length- N sequence $x[n]$:

$$x[\langle -n \rangle_N] = \begin{cases} x[0], & \text{for } n = 0 \\ x[N - n], & \text{for } 1 \leq n \leq N - 1 \end{cases}$$



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Circular convolution



- To develop a convolution-like operation resulting in a length- N sequence $y_C[n]$, we need to apply circular time-reversal and circular time-shift operations.
- **Circular convolution** defined as:

$$y_C[n] = \sum_{m=0}^{N-1} g[m]h[\langle n - m \rangle_N]$$
$$= \mathbf{g[n]} \circledast^N \mathbf{h[n]} = \mathbf{g[n]} \circledast^N \mathbf{h[n]}$$

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- The N-point circular convolution can be written in matrix form as:



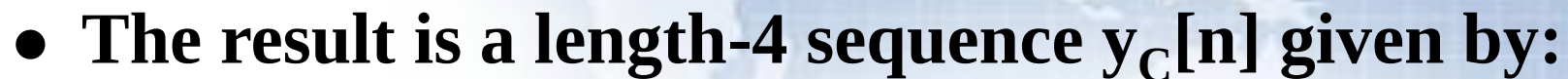
$$\begin{bmatrix} y_C[0] \\ y_C[1] \\ y_C[2] \\ \vdots \\ y_C[N-1] \end{bmatrix} = \begin{bmatrix} h[0] & h[N-1] & h[N-2] & \cdots & h[1] \\ h[1] & h[0] & h[N-1] & \cdots & h[2] \\ h[2] & h[1] & h[0] & \cdots & h[3] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h[N-1] & h[N-2] & h[N-3] & \cdots & h[0] \end{bmatrix} \begin{bmatrix} g[0] \\ g[1] \\ g[2] \\ \vdots \\ g[N-1] \end{bmatrix}$$

- The elements of each diagonal of the $N \times N$ matrix are equal
- Such a matrix is called a **circulant matrix**

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Logo of the University of Electronic Science and Technology of China (UESTC), established in 1956. The logo features a blue circle with the university's name in Chinese characters (电子科技大学) around the top and the year 1956 at the bottom. In the center, the acronym UESTC is displayed in white, with an orange and blue orbital design behind it.

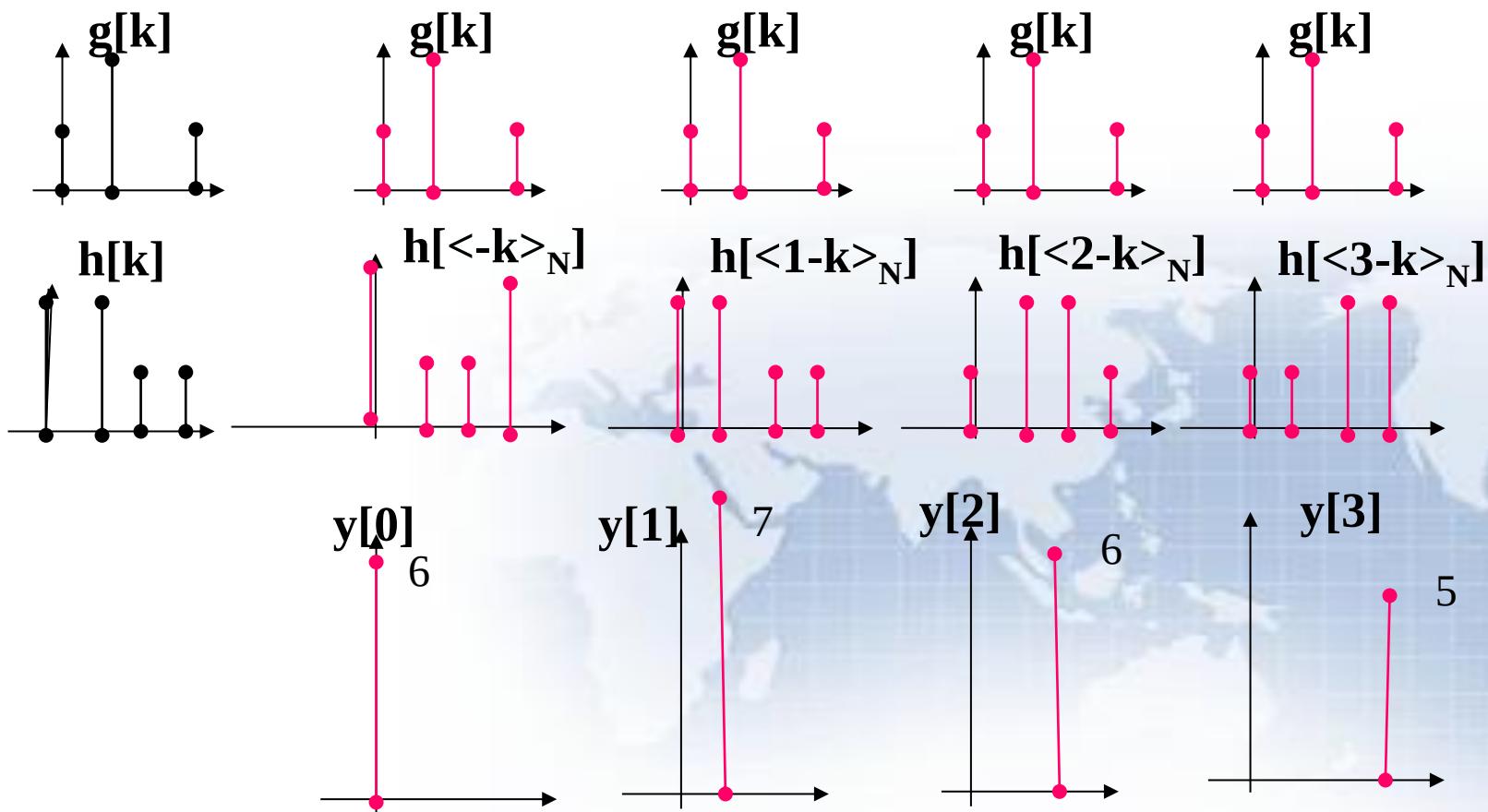
as sketched below:



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$$\{g[n]\} = \{1 \quad 2 \quad 0 \quad 1\}, \quad \{h[n]\} = \{2 \quad 2 \quad 1 \quad 1\}$$

$\uparrow$
 $\uparrow$



$$y_c[n] = 6\delta[n] + 7\delta[n-1] + 6\delta[n-2] + 5\delta[n-3]$$

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From the above we observe:



$$\begin{aligned}y_C[0] &= \sum_{m=0}^3 g[m] h[\langle -m \rangle_4] \\&= g[0]h[0] + g[1]h[3] + g[2]h[2] + g[3]h[1] \\&= (1 \times 2) + (2 \times 1) + (0 \times 1) + (1 \times 2) = 6 \\y_C[1] &= \sum_{m=0}^3 g[m] h[\langle 1-m \rangle_4] \\&= g[0]h[1] + g[1]h[0] + g[2]h[3] + g[3]h[2] \\&= (1 \times 2) + (2 \times 2) + (0 \times 1) + (1 \times 1) = 7\end{aligned}$$



$$\begin{aligned}y_C[2] &= \sum_{m=0}^3 g[m] h[\langle 2-m \rangle_4] \\&= g[0]h[2] + g[1]h[1] + g[2]h[0] + g[3]h[3] \\&= (1 \times 1) + (2 \times 2) + (0 \times 2) + (1 \times 1) = 6\end{aligned}$$

$$\begin{aligned}y_C[3] &= \sum_{m=0}^3 g[m] h[\langle 3-m \rangle_4] \\&= g[0]h[3] + g[1]h[2] + g[2]h[1] + g[3]h[0] \\&= (1 \times 1) + (2 \times 1) + (0 \times 2) + (1 \times 2) = 5\end{aligned}$$

$$\mathbf{y[n]=6\delta[n]+7\delta[n-1]+6\delta[n-2]+5\delta[n-3]}$$

Tabular Method in Circular convolution



Example: We want to obtain the convolution sum $y_c[n]$ by the following step: $y_c[n]=g[n] \textcircled{*} h[n]$

n:	0	1	2	3	4	5
g[n]:	g[0]	g[1]	g[2]			
h[n]:	h[0]	h[1]	h[2]			
	g[0]h[0]	g[1]h[0]	g[2]h[0]			
		g[2]h[1]	g[0]h[1]	g[1]h[1]	g[2]h[1]	
			g[1]h[2]	g[2]h[2]	g[0]h[2]	g[1]h[2]
y _c [n]:	y _c [0]	y _c [1]	y _c [2]			

MOOC

The comparison of linear convolution & circular convolution



- **The circular convolution:**

$$y_C[n] = \sum_{m=0}^{N-1} g[m]h[\langle n - m \rangle_N]$$

- **The linear convolution:**

$$y_L[n] = \sum_{m=0}^{N-1} g[m]h[n - m]$$

Please look at Fig5.6

5.5 Classifications of Finite-Length Sequences



- The **circular conjugate-symmetric** sequence and **circular conjugate-antisymmetric** sequence;
- The **geometric symmetric** sequence and **geometric antisymmetric** sequence;
- Four types of geometric symmetry sequences.

5.5.1 Classification Based on Conjugate Symmetry



- **circular conjugate-symmetry:**

$$x[n] = x^*[\langle -n \rangle_N] = x^*[\langle N - n \rangle_N], 0 \leq n \leq N - 1$$

- **circular conjugate-antisymmetry:**

$$x[n] = -x^*[\langle -n \rangle_N] = -x^*[\langle N - n \rangle_N], 0 \leq n \leq N - 1$$

MOOC



- A complex length- N sequence $x[n]$ can be expressed as:

$$x[n] = x_{cs}[n] + x_{ca}[n], 0 \leq n \leq N-1$$

where, $x_{cs}[n]$ is its **circular conjugate-symmetric part**, and $x_{ca}[n]$ is its **circular conjugate-antisymmetric part**, defined by:

$$x_{cs}[n] = \frac{1}{2} (x[n] + x^*[\langle -n \rangle_N]), 0 \leq n \leq N-1$$

$$x_{ca}[n] = \frac{1}{2} (x[n] - x^*[\langle -n \rangle_N]), 0 \leq n \leq N-1$$

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- For a real sequence $x[n]$, it can be expressed as:

$$x[n] = x_{ev}[n] + x_{od}[n], 0 \leq n \leq N-1$$

Where, $x_{ev}[n]$ is its **circular even part**, and $x_{ca}[n]$ is its **circular odd part**, defined by:

$$x_{ev}[n] = \frac{1}{2} (x[n] + x[\langle -n \rangle_N])$$

$$x_{od}[n] = \frac{1}{2} (x[n] - x[\langle -n \rangle_N])$$

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5.5.2 Classification Based on Geometric Symmetry



- Two types of geometric symmetries are usually defined:

(1) Symmetric sequence:

$$x[n] = x[N - 1 - n]$$

(2) Antisymmetric sequence:

$$x[n] = -x[N - 1 - n]$$

Since the length N of a sequence can be either even or odd, four types of geometric symmetry are defined.

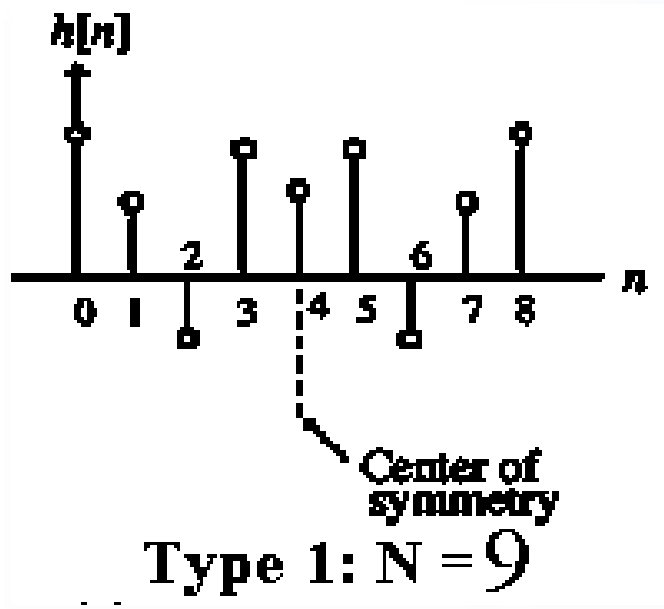
Four types of geometric symmetry sequences



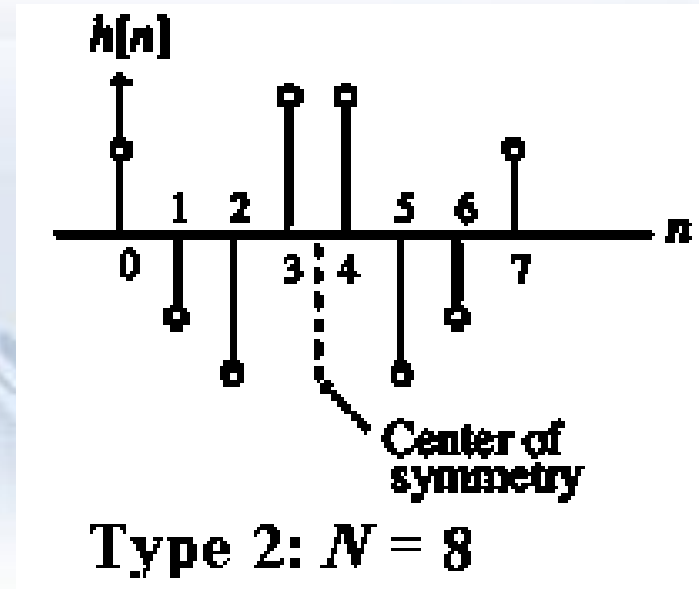
- **Type1:** Symmetric impulse response with odd length.
- **Type2:** Symmetric impulse response with even length.
- **Type3:** Antisymmetric impulse response with odd length.
- **Type4:** Antisymmetric impulse response with even length.

● In the general case for Type1 & Type2 FIR filters, the phase function is of the form:

$$\theta(\omega) = -\frac{N-1}{2}\omega + \beta$$



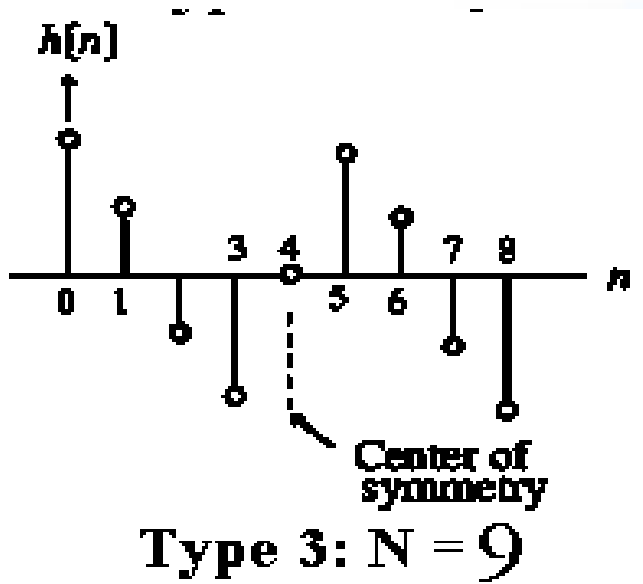
$$\theta(\omega) = -4\omega + \beta$$



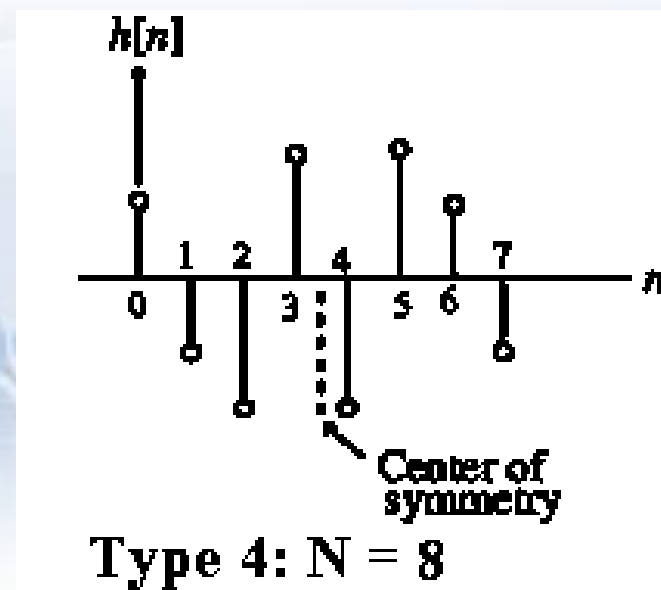
$$\theta(\omega) = -\frac{7}{2}\omega + \beta$$

● In the general case for Type3 & Type4 FIR filters, the phase function is of the form:

$$\theta(\omega) = -\frac{N-1}{2}\omega + \frac{\pi}{2} + \beta$$



$$\theta(\omega) = -4\omega + \frac{\pi}{2} + \beta$$



$$\theta(\omega) = -\frac{7}{2}\omega + \frac{\pi}{2} + \beta$$

5.6 DFT Properties

Table 5.1: Symmetric Relations 1



Length-N Sequence	N-point DFT
$x[n]$	$X[k]$
$x^*[n]$	$X^*[\langle -k \rangle_N]$
$x^*[\langle -n \rangle_N]$	$X^*[k]$
$\text{Re}\{x[n]\}$	$X_{cs}[k] = \{X[\langle k \rangle_N] + X^*[\langle -k \rangle_N]\}/2$
$j\text{Im}\{x[n]\}$	$X_{ca}[k] = \{X[\langle k \rangle_N] - X^*[\langle -k \rangle_N]\}/2$
$x_{cs}[n]$	$\text{Re}\{X[k]\}$
$x_{ca}[n]$	$j\text{Im}\{X[k]\}$

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Note: $x[n]$ is a **complex** sequence. $x_{cs}[n]$ and $x_{ca}[n]$ are the circular conjugate-symmetric and antisymmetric parts of $x[n]$, respectively.

Table 5.2: Symmetric Relations 2



Length-N Sequence	N-point DFT
Real $x[n]$	$X[k] = \text{Re}\{X[k]\} + j\text{Im}\{X[k]\}$
$x_{\text{ev}}[n]$	$\text{Re}\{X[k]\}$
$x_{\text{od}}[n]$	$j\text{Im}\{X[k]\}$
Symmetry relations	$X[k] = X^*[\langle -k \rangle_N]$
	$\text{Re}X[k] = \text{Re}X[\langle -k \rangle_N]$
	$\text{Im}X[k] = -\text{Im}X[\langle -k \rangle_N]$
	$ X[k] = X[\langle -k \rangle_N] $
	$\arg X[k] = -\arg X[\langle -k \rangle_N]$

Note: $x[n]$ is a **real** sequence. $x_{\text{ev}}[n]$ and $x_{\text{od}}[n]$ are the circular even and odd parts of $x[n]$, respectively.

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Example: Consider the length-14 **real** sequence

$x[n]$ has 14-point DFT $X[k]$.

If $X[2] = -1 + j3$

Then $X[12] = X^*[2] = -1 - j3$

$$X[12] = X[\langle -2 \rangle_N] = X[\langle N - 2 \rangle_N]$$

(N=14)

MOOC

5.7 DFT Theorems



Type of Property	length-N sequence	N-point DFT
	$g[n]$	$G[k]$
	$h[n]$	$H[k]$
Linearity	$ag[n]+bh[n]$	$aG[k]+bH[k]$
Circular Time-shifting	$g[\langle n-n_0 \rangle_N]$	$W_N^{kn_0}G[k]$
Frequency-shifting	$W_N^{-k_0n}g[n]$	$G[\langle k-k_0 \rangle_N]$
Duality	$G[n]$	$N[g\langle -k \rangle_N]$
Circular Convolution	$\sum_{m=0}^{N-1} g[m]h[\langle n-m \rangle_N]$	$G[k]H[k]$
Modulation	$g[n]h[n]$	$\frac{1}{N} \sum_{m=0}^{N-1} G[m]H[\langle k-m \rangle_N]$
Parseval's relation	$\sum_{n=0}^{N-1} x[n] ^2 = \frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$	

MOOC

Example: Consider the length- N sequence



$$y[n] = \begin{cases} 1, & n = m \\ 0, & 0 \leq n \leq m-1, m+1 \leq n \leq N-1 \end{cases}$$

$$\therefore x[n] = \begin{cases} 1, & n = 0 \\ 0, & 1 \leq n \leq N-1 \end{cases}, X[k] = 1, 0 \leq k \leq N-1$$

$$\text{and } y[n] = x[\langle n - m \rangle_N]$$

$$\therefore Y[k] = X[k]W_N^{km} = W_N^{km}, 0 \leq k \leq N-1$$

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Example - Consider the two length-4 sequences:

$$\{g[n]\} = \{1 \quad 2 \quad 0 \quad 1\}, \quad \{h[n]\} = \{2 \quad 2 \quad 1 \quad 1\}$$

$\uparrow \qquad \qquad \qquad \uparrow$

$$y_c[n] = g[n] \textcircled{4} h[n], \quad 0 \leq n \leq 3$$

- **We can use DFT to compute the $y_c[n]$.**
- **The two 4-point DFTs of $g[n]$ and $h[n]$ can be computed using the matrix relation.**

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$$\begin{bmatrix} G[0] \\ G[1] \\ G[2] \\ G[3] \end{bmatrix} = D_4 \begin{bmatrix} g[0] \\ g[1] \\ g[2] \\ g[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1-j \\ -2 \\ 1+j \end{bmatrix}$$

$$\begin{bmatrix} H[0] \\ H[1] \\ H[2] \\ H[3] \end{bmatrix} = D_4 \begin{bmatrix} h[0] \\ h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 1-j \\ 0 \\ 1+j \end{bmatrix}$$

D_4 is the 4-point DFT matrix.

MOOC

- If $Y_C[k]$ denotes the 4-point DFT of $y_C[n]$, then:

$$Y_C[k] = G[k]H[k], 0 \leq k \leq 3$$

- Thus:

$$\begin{bmatrix} Y_C[0] \\ Y_C[1] \\ Y_C[2] \\ Y_C[3] \end{bmatrix} = \begin{bmatrix} G[0]H[0] \\ G[1]H[1] \\ G[2]H[2] \\ G[3]H[3] \end{bmatrix} = \begin{bmatrix} 24 \\ -j2 \\ 0 \\ j2 \end{bmatrix}$$

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- So $y_C[n]$ is 4-point IDFT of $Y_C[k]$:

$$\begin{bmatrix} y_C[0] \\ y_C[1] \\ y_C[2] \\ y_C[3] \end{bmatrix} = \frac{1}{4} D_4^* \begin{bmatrix} Y_C[0] \\ Y_C[1] \\ Y_C[2] \\ Y_C[3] \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 24 \\ -j2 \\ 0 \\ j2 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \\ 6 \\ 5 \end{bmatrix}$$

MOOC

5.8 Fourier-Domain Filtering



- Often one is interested in removing the components of a finite-length discrete-time signal in one or more frequency bands.
- A **straightforward approach** to remove the unwanted components from a signal is to implement the **filtering in the Fourier domain**.
- The Fourier-domain filtering is usually implemented using DFT. However, the DFT-based filtering will always lead to some small ripples in the filtered response.

Please look at P232, Fig5.10

Fourier-Domain Filtering



Based on DTFT convolution theorem:

$$\sum_{m=0}^{N-1} g[m]h[n-m] \longleftrightarrow G(e^{j\omega})H(e^{j\omega})$$

- In practical condition, $g[n]$ and $h[n]$ are **real-valued signal**, so the **imaginary parts** of the samples of the IDTFT of the product of their FT will be theoretically all **zeros**.
- In practice, the imaginary parts are very small numbers due to computational errors.
- The real part of the IDTFT of the product -- $y[n]$ is kept as the filtered response.

MOOC

5.9 Computation of the DFT of Real sequences



- In most practical applications, sequences are real.
- In DFT definition, the sequence is assumed to be complex.
- The DFTs of two N -point real sequences can be computed from a length- N complex sequences.
- And a $2N$ -point DFT of a length- $2N$ real sequence can be determined from a length- N complex sequences.



5.9.1 *N*-point DFTs of two real sequence using a single *N*-point DFT

- Let $g[n]$ and $h[n]$ be two real sequences of length N each, with $G[k]$ and $H[k]$ denoting their N -point DFTs.

$$x[n] = g[n] + jh[n]$$

- We build:

$$H[k] = \frac{1}{2j} \{X[k] - X^*[\langle -k \rangle_N]\}$$

- Then, we have:

$$G[k] = \frac{1}{2} \{X[k] + X^*[\langle -k \rangle_N]\}$$

5.9.2 $2N$ -Point DFT of a Real Sequences Using a Single N -Point DFT



- Let $v[n]$ be a real sequence of length $2N$ with $V[k]$ denoting its $2N$ -point DFT.
- We can define two real sequences $g[n]$ and $h[n]$ of length N each as:

$$g[n] = v[2n], h[n] = v[2n + 1]$$

- So,

$$V[k] = G[\langle k \rangle_N] + W_{2N}^k H[\langle k \rangle_N]$$

5.10 Linear Convolution Using DFT



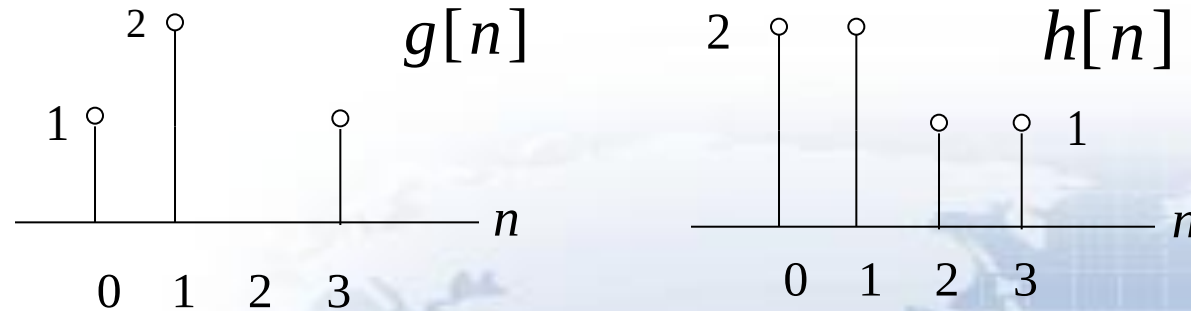
- Linear convolution is a key operation in many signal processing applications.
- Since a DFT can be efficiently implemented using FFT algorithms, it is of interest to develop methods for the implementation of linear convolution using the DFT.
- How to do?
 - Linear Convolution of Two Finite-Length Sequences;
 - Linear Convolution of a Finite-Length Sequence with an Infinite-Length Sequence.

MOOC

5.10.1 Linear Convolution of Two Finite-Length Sequences



Example: Given two length-4 sequences:



Let's extended the two length-4 sequences to length 7 by appending each with three zero-valued samples, i.e.

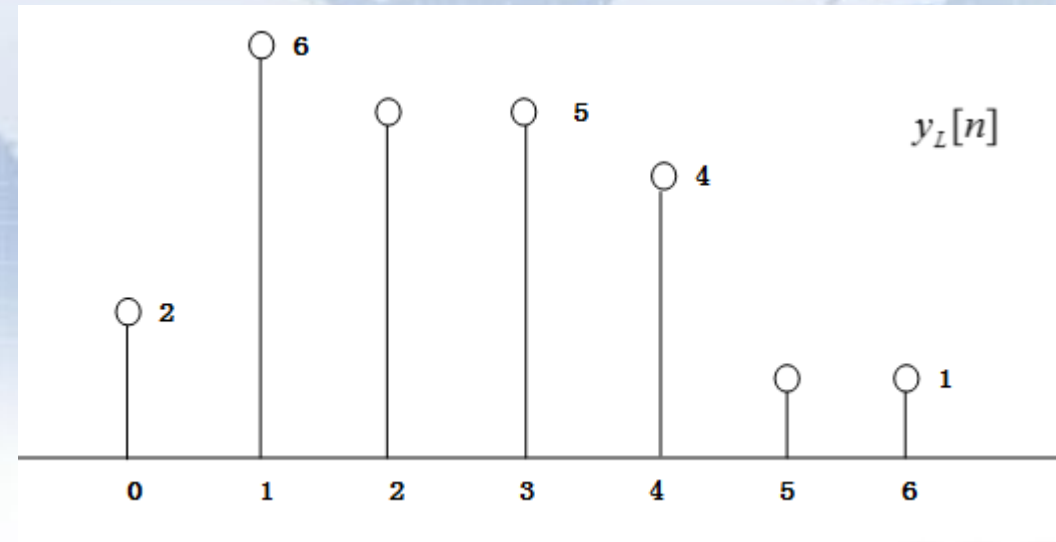
$$g_e[n] = \begin{cases} g[n], & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 6 \end{cases} \quad h_e[n] = \begin{cases} h[n], & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 6 \end{cases}$$

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- Now determine the 7-point circular convolution of $g_e[n]$ and $h_e[n]$:

$$y_c[n] = \sum_{m=0}^6 g_e[m] h_e[\langle n - m \rangle_7], \quad 0 \leq n \leq 6$$

- As can be seen: $y_c[n]$ is same as the sequence $y_L[n]$, which is the linear convolution of $g[n]$ and $h[n]$.



- Let $g[n]$ and $h[n]$ be two finite-length sequences of length N and M , respectively.
- Define two length- L ($L=N+M-1$) sequences:

$$g_e[n] = \begin{cases} g[n], & 0 \leq n \leq N-1 \\ 0, & N \leq n \leq L-1 \end{cases}$$

$$h_e[n] = \begin{cases} h[n], & 0 \leq n \leq M-1 \\ 0, & M \leq n \leq L-1 \end{cases}$$

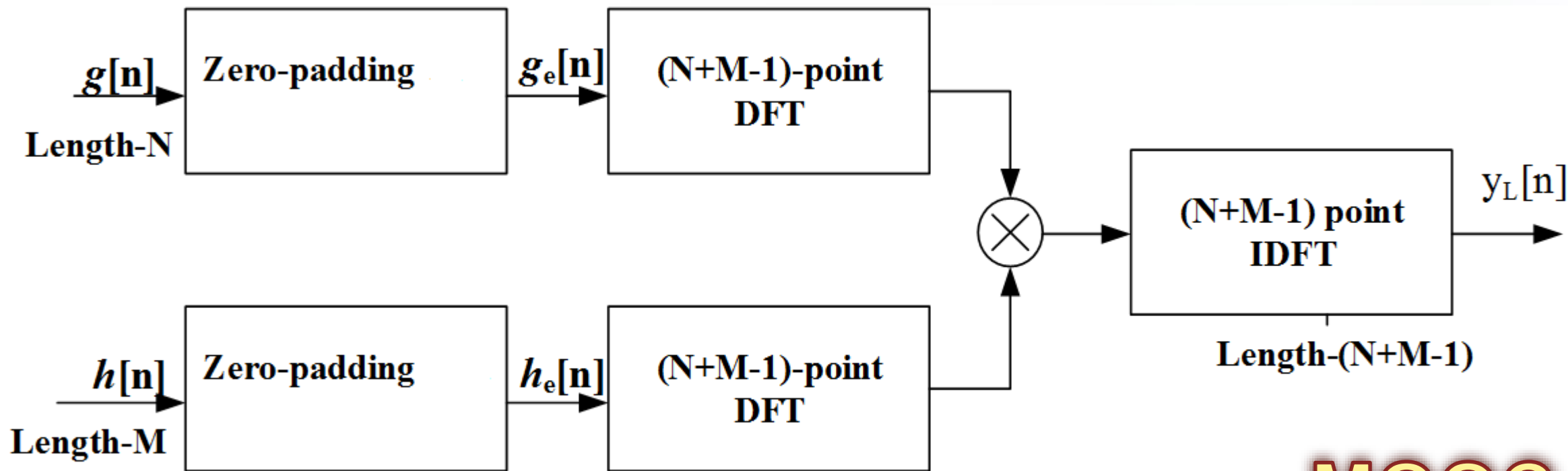
- Then $y_L[n] = g[n]*h[n] = g_e[n] \textcircled{L} h_e[n]$

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Linear Convolution Using DFT



- The corresponding implementation scheme is:



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5.10.2 The Cyclic Prefix



- In some applications, it is required to compute only a portion of the sequence $\{y[n]\}$ of length N .
- Using an N -point DFT and IDFT by appending the longer sequence with a subsequence called the **cyclic prefix**.
- The cyclic prefix plays an important role in multicarrier-based digital communication.

Example 5.18

5.10.3 Linear Convolution of a Finite-Length Sequence with an Infinite-Length Sequence



- Consider the DFT-based implementation of:

$$y[n] = \sum_{\ell=0}^{M-1} h[\ell] x[n-\ell] = h[n] \circledast x[n]$$

where $h[n]$ is a finite-length sequence of length M and $x[n]$ is an **infinite** length (or a finite length sequence of length much greater than M).

- There are two methods to solve this problem:
 - Overlap-Add
 - Overlap-Save

Overlap-Add Method



- Segment $x[n]$ into a set of contiguous finite-length subsequences $x_m[n]$ of length N each:

$$x[n] = \sum_{m=0}^{\infty} x_m[n - mN] \quad x_m[n] = \begin{cases} x[n + mN], & 0 \leq n \leq N - 1 \\ 0, & \text{otherwise} \end{cases}$$

- **Then:**

$$y[n] = h[n] \circledast x[n] = \sum_{m=0}^{\infty} y_m[n - mN]$$

where $y_m[n] = h[n] \circledast x_m[n]$

Overlap-Add Method

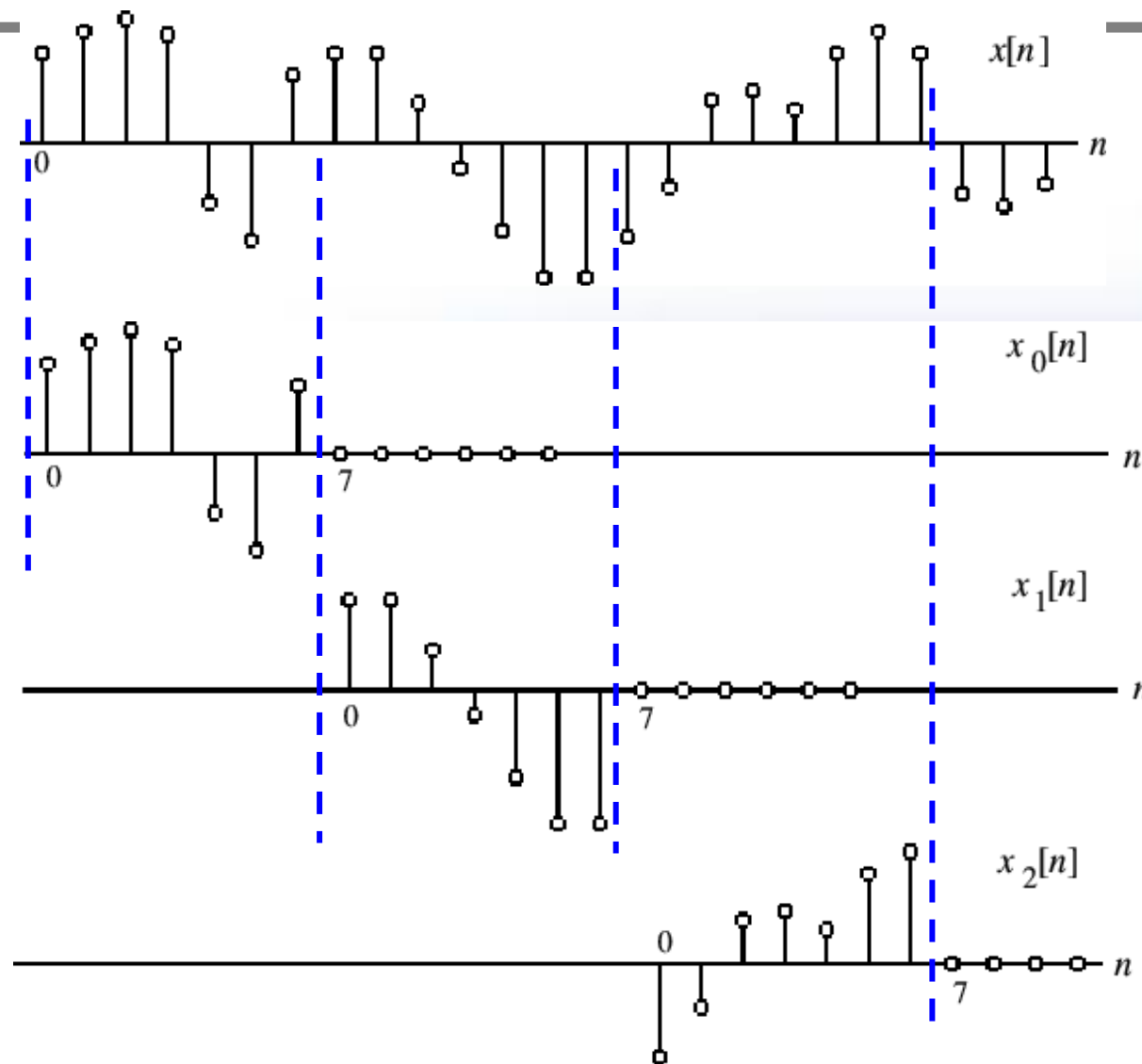


- As a result, the desired linear convolution $y[n] = h[n] \circledast x[n]$ has been broken up into a sum of infinite number of **short-length linear convolutions** of length **$N+M-1$** each:

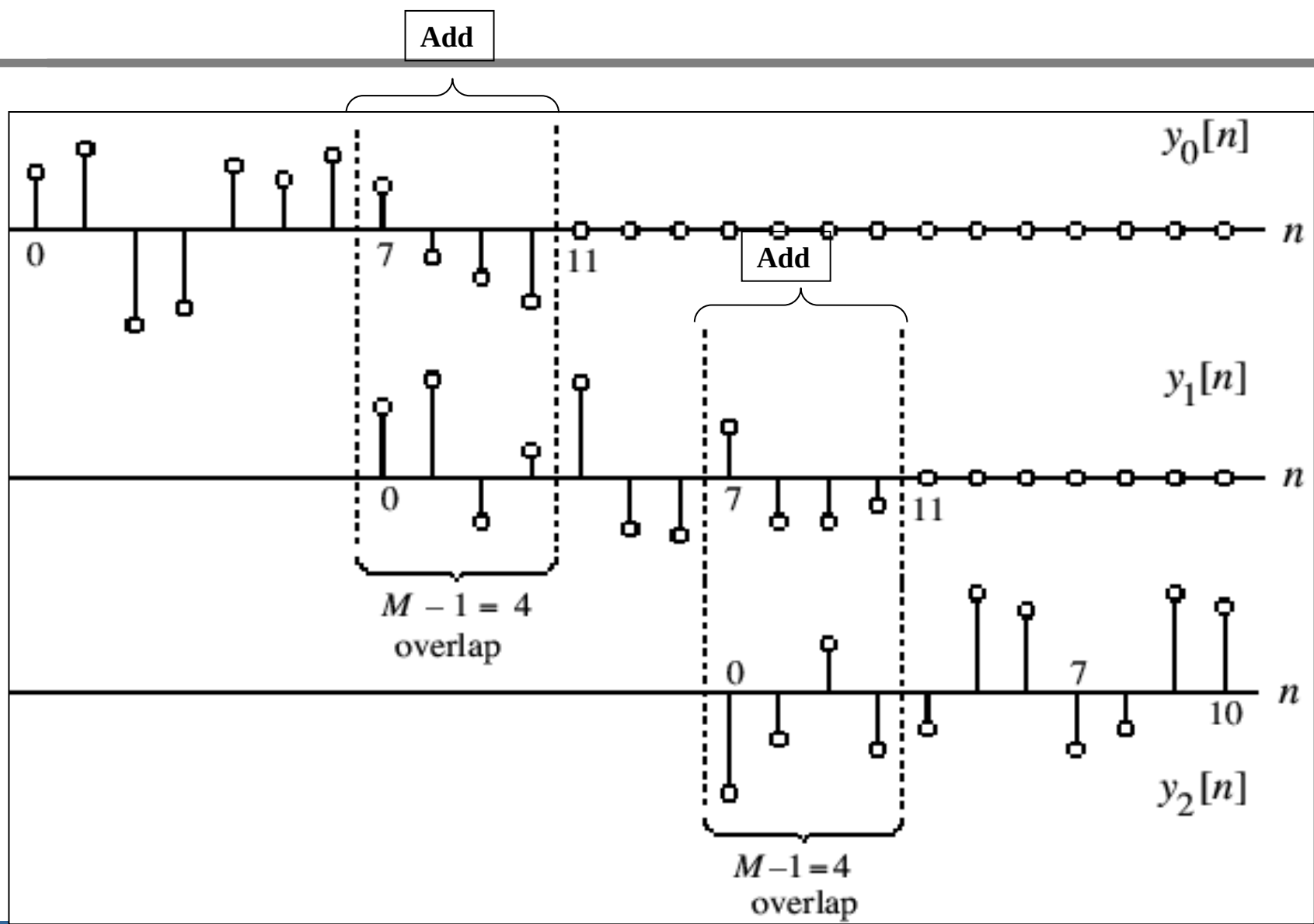
$$y_m[n] = x_m[n] \circledast h[n]$$

- These short convolutions can be implemented using the DFT-based approach.
- There is an **overlap of $M - 1$ samples** between these two short linear convolutions.

Segment $x[n]$ into subsequences



This figure illustrate Overlap-Add method for M = 5 and N = 7



$$\begin{aligned}
 y[n] &= y_0[n], & 0 \leq n \leq 6 \\
 y[n] &= y_0[n] + y_1[n-7], & 7 \leq n \leq 10 \\
 y[n] &= y_1[n-7], & 11 \leq n \leq 13 \\
 y[n] &= y_1[n-7] + y_2[n-14], & 14 \leq n \leq 17 \\
 y[n] &= y_2[n-14], & 18 \leq n \leq 20 \\
 &\vdots
 \end{aligned}$$

Overlap-Add Method



- The above procedure is called the overlap-add method since the results of the short linear convolutions overlap and the overlapped portions are added to get the correct final result.
- The function **fftfilt** can be used to implement the above method.
- Program 5_5 illustrates the use of **fftfilt** in the filtering of a noise-corrupted signal using a length-3 moving average filter.

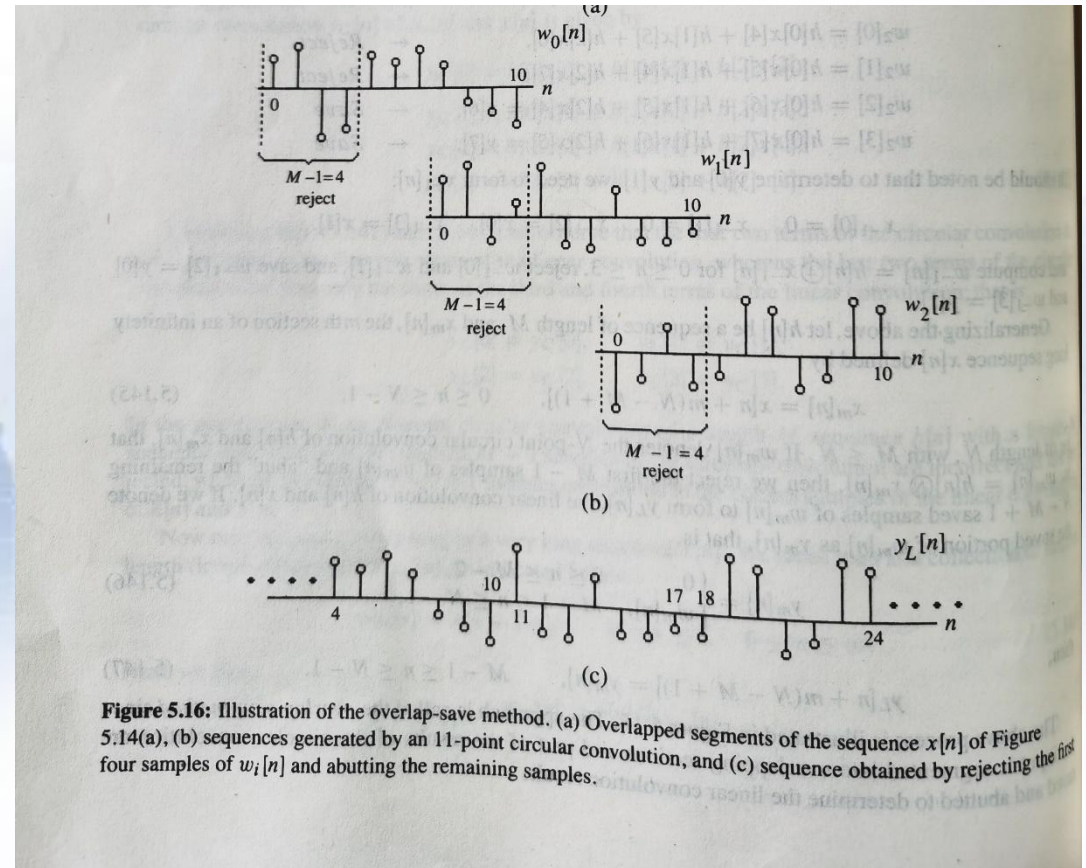
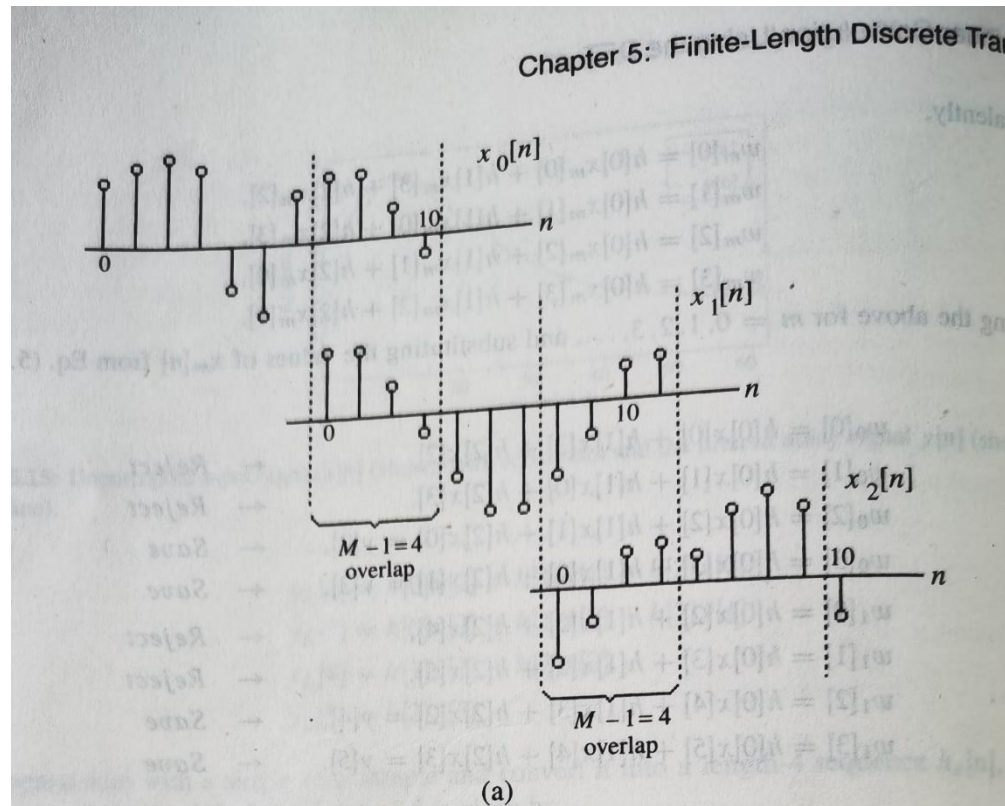
Overlap-Save Method



- At first, to segment $x[n]$ into **overlapping blocks** $x_m[n]$.
- Keeping the terms of the **circular convolution** of $h[n]$ with $x_m[n]$ that corresponds to the terms obtained by a linear convolution of $h[n]$ and $x_m[n]$.
- Throwing away the other part of the circular convolution.

Please look at P244, Fig5.16

Overlap-Save Method



Other Transforms



5.11 STFT

5.12 DCT

5.13 Haar Transform

Homework



- 5.4, 5.8 (a) (b), 5.13, 5.20, 5.25, 5.32, 5.36, 5.39, 5.40, 5.53, 5.64, 5.69, 5.76
- M5.2, M5.10, M5.11